# Deep Q-Networks

**MLDS 490 - Assignment 2**
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Repository: **mlds490-hw2**
https://github.com/ZikunZheng1129/mlds490-hw2

## Scope and measured results

This assignment implements standard DQN separately for CartPole-v1 and ALE/MsPacman-v5. Both agents use a neural action-value function, uniform replay, a periodically copied target network and epsilon-greedy exploration. All figures and statistics below are generated from the accompanying saved records.

| Measure | CartPole-v1 | MsPacman-v5 |
|---|---|---|
| Training episodes | 1500 | 2000 |
| Environment decisions | 155,667 | 1,303,878 |
| Selected checkpoint episode | 1300 | 1500 |
| Frozen evaluation episodes | 500 | 500 |
| Mean raw reward | 500.00 | 1420.52 |
| Sample standard deviation | 0.00 | 511.69 |

**Interpretation.** CartPole is judged by survival to its 500-step time limit. MsPacman has no score threshold in the assignment; improvement is assessed using training curves and frozen-policy comparisons to untrained and random policies.

**Evidence provided.** Each task includes maximum-Q and reward curves, a 500-episode reward histogram, a task-specific nonstandard visualization, controlled validation experiments, edge-case discussion and an auditable ten-step trace. The final pages document how to reproduce the work and the experiments proposed for three additional months.

**Scope of claims.** Evaluation episodes measure variation across initial seeds and environment stochasticity. A single training seed per selected run does not establish robustness across independent retraining runs.

# Method, environments and hyperparameters

For a sampled transition, the regression target is **y = r + gamma (1 - terminated) max Q_target(next state, action)**. The online network minimizes mean Huber loss against this detached target. Adam updates only the online parameters; gradient norm is clipped at 10. Target parameters are copied after the specified number of optimizer updates.

| Setting | CartPole-v1 | MsPacman-v5 |
|---|---|---|
| Discount factor | 0.95 | 0.99 |
| Adam learning rate | 0.0005 | 0.0001 |
| Batch size | 64 | 32 |
| Replay capacity | 50000 | 50000 |
| Warm-up decisions | 1000 | 10000 |
| Decisions per update | 1 | 4 |
| Optimizer updates per target copy | 250 | 2000 |
| Linear exploration-decay decisions | 30000 | 500000 |
| Final training epsilon | 0.02 | 0.05 |
| Training seed | 7 | 7 |

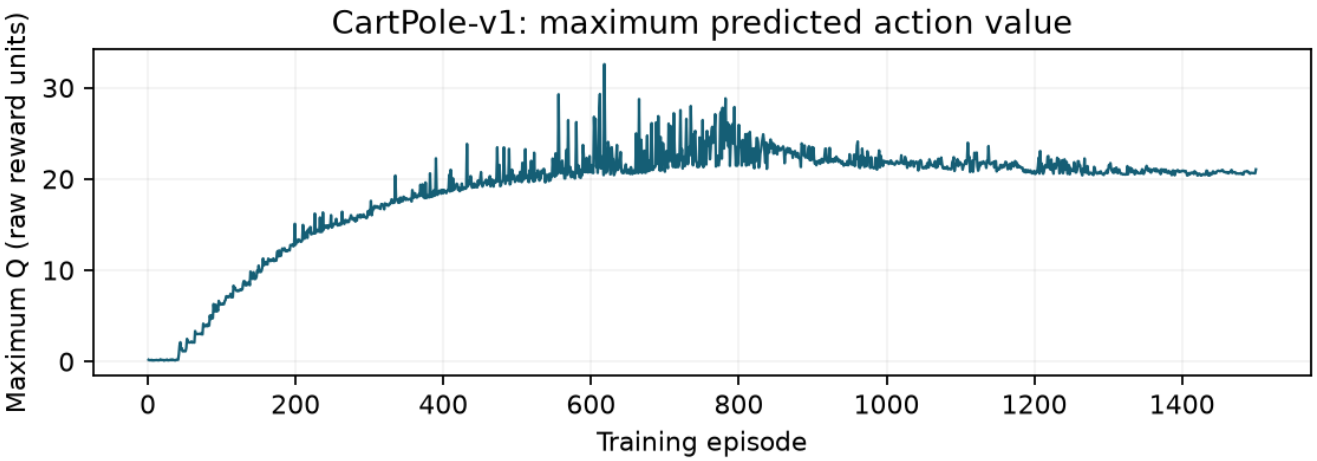
**Networks.** CartPole uses 4 inputs, two 128-unit ReLU hidden layers and 2 raw Q outputs. MsPacman uses four 88 x 80 grayscale frames; convolutions are 32 channels (8 x 8, stride 4), 64 (4 x 4, stride 2), 64 (3 x 3, stride 1), followed by 256 ReLU units and 9 raw Q outputs. There is no policy softmax.

**Image and reward handling.** The assignment's crop, channel averaging and contrast mask are used. Frames are stored as uint8; the network computes (pixel - 128)/128. Frame stacking provides short motion history. MsPacman learning uses sign-clipped rewards; all reported game scores and evaluation rewards are unmodified. CartPole rewards are unmodified.

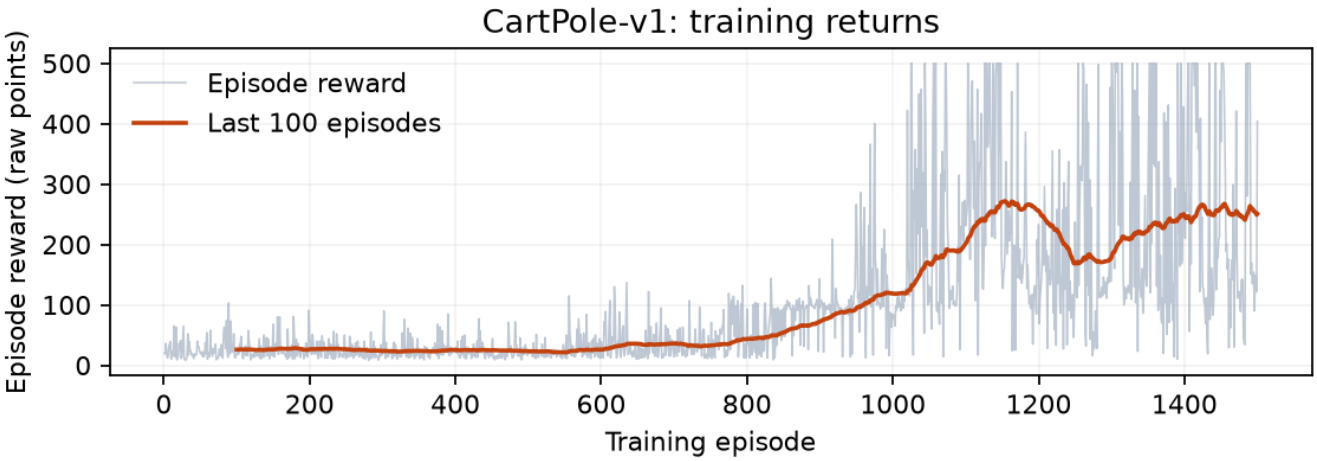
**Environment fidelity.** MsPacman is ALE/MsPacman-v5 with nine actions, frame skip 4 and sticky-action probability 0.25. A life loss does not end a reported game. CartPole retains the standard 500-step limit. Both termination and truncation stop a rollout, but only true termination suppresses the next-state Bellman term. [1-4]

**Checkpoint selection.** Development evaluations use a fixed seed range starting at 40000 and greedy actions. The highest development mean selects policy.pt. The independent 500 episodes start at 100000 + training seed and never select the checkpoint. Initial and latest checkpoints are retained. CartPole may stop after two consecutive development checks in which every episode reaches 500 steps.

## CartPole: training behavior




**Maximum Q.** Each point is the maximum over all visited states and all action outputs in that training episode. The online network changes during the episode. Q values are predictions, not measured game scores; a larger Q alone does not prove learning.

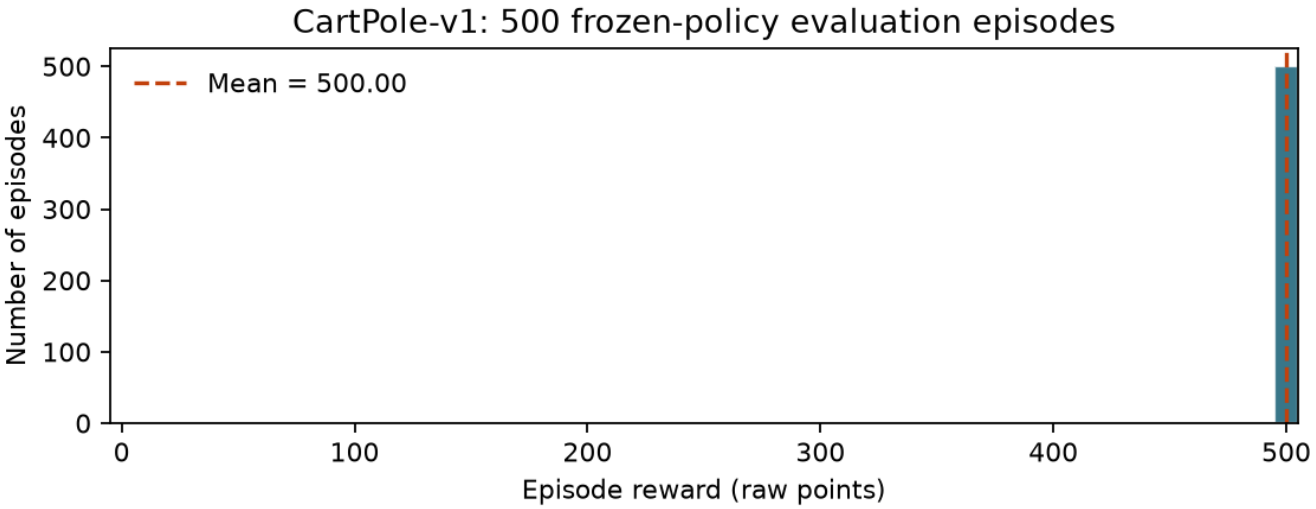



The first and final 100 training episodes average **25.98** and **250.68** raw reward, respectively (difference +224.70). The 100-episode moving average begins only when 100 complete episodes exist. Exploration changes over training, so the fixed-policy held-out comparisons on the next page provide a separate check of actual learned behavior.

Frozen evaluation uses the development-selected checkpoint from episode **1300**. The training curve includes exploratory actions and subsequent parameter updates; its final moving average therefore need not match the selected checkpoint's greedy evaluation score.

The largest predicted Q is **32.58**. With reward 1 per step and gamma=0.95, discounted return cannot exceed **20.00**. The excess indicates Q-value overestimation or function-approximation error; Q growth is not a correctness certificate.

## CartPole: frozen evaluation and learning validation




**Required 500-episode result:** mean 500.00; sample standard deviation 0.00; range [500, 500]. Evaluation uses epsilon=0 and seeds 100007 through 100506; no updates or replay sampling occur. Ending flags: 0 true terminations and 500 time truncations.

| Held-out policy | Episodes | Mean reward | Sample SD |
|---|---|---|---|
| Uniform random (epsilon=1) | 50 | 22.46 | 12.29 |
| Untrained network (epsilon=0) | 50 | 9.34 | 0.66 |
| Selected network (epsilon=0) | 50 | 500.00 | 0.00 |

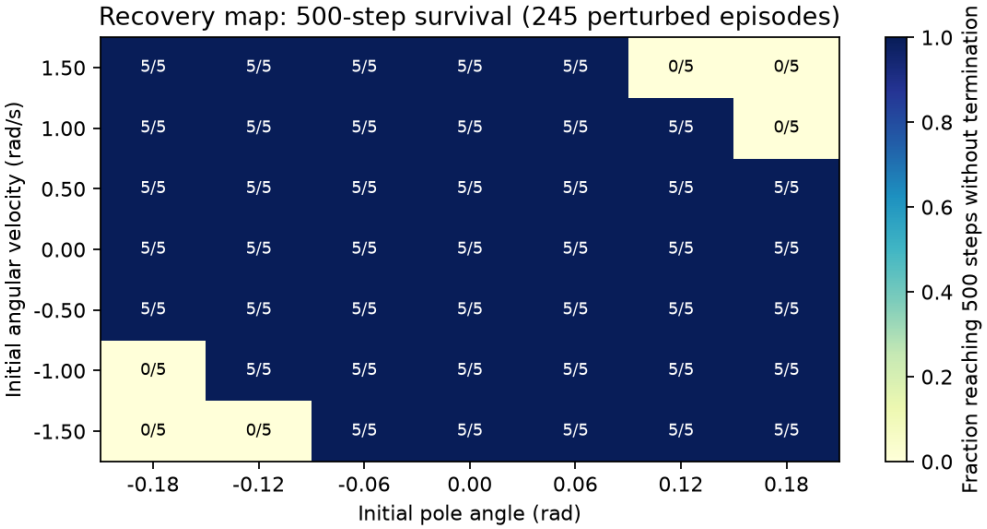
**Controlled learning check.** The three policies use the same 50 held-out reset seeds starting at 80000, separate from checkpoint selection and the required 500-episode evaluation. Initial and selected networks use identical greedy evaluation settings; the uniform-random control uses epsilon=1. Matched seeds do not imply identical trajectories once actions differ.

Selected minus random: mean paired difference 477.54 points; 95% paired-bootstrap interval [473.82, 480.68] (n=50 pairs). The paired interval supports a positive improvement under this evaluation protocol.

Selected minus initial: mean paired difference 490.66 points; 95% paired-bootstrap interval [490.48, 490.84] (n=50 pairs). The paired interval supports a positive improvement under this evaluation protocol.

**500-step objective:** 500/500 episodes (100.0%) survived to the time limit without true termination. Every evaluation episode reached the objective.

# CartPole: task-specific diagnostic and edge cases



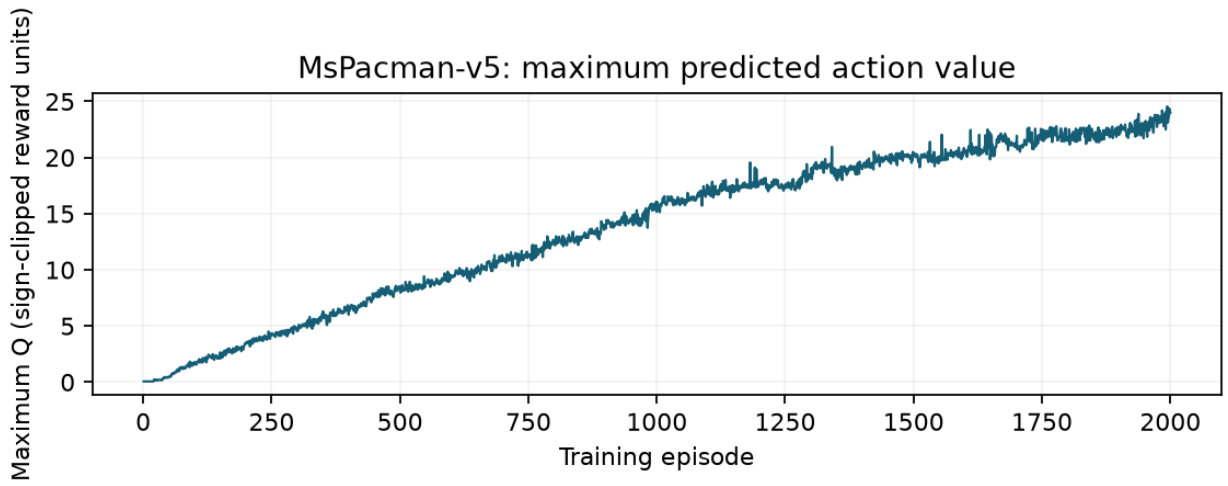

**Nonstandard recovery map.** Each cell reports successes/trials for deliberately perturbed starting angle and angular velocity, with small independent reset jitters. Across 245 trials, survival is 87.8% and mean length is 439.1 steps. Failure near difficult initial states reveals the policy's recovery boundary; no monotonic pattern is assumed. These changed resets do not replace the standard 500-episode evaluation.

**Auditable first ten steps.** The selected policy's validation rollout at seed 80000 records observations (or image summaries/hashes), every action value, selected action, raw/training rewards, target and ending flags in validation_first10_steps.json.

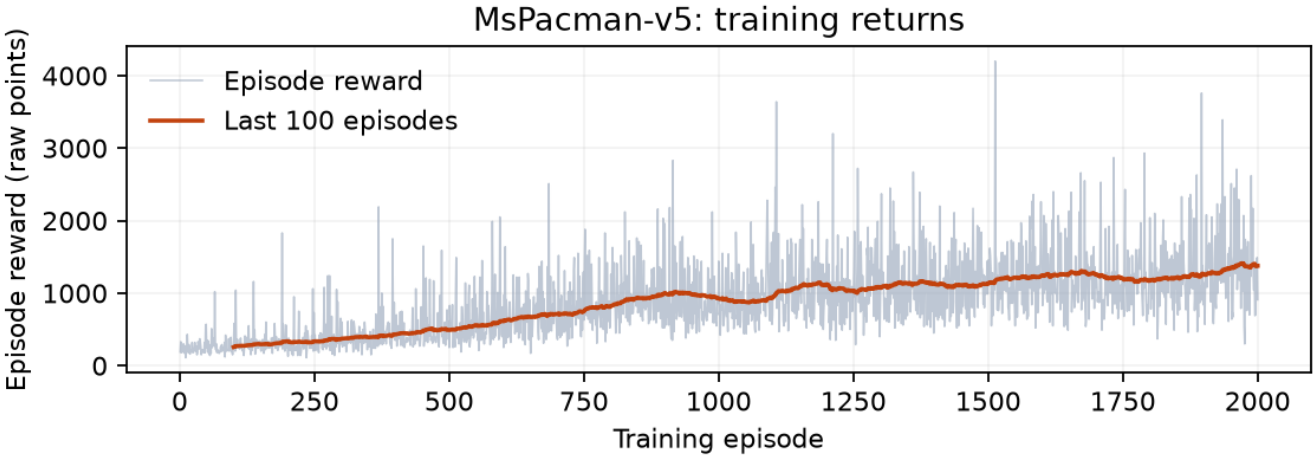
| Step | Action | Raw reward | Training reward | Target | Term / trunc |
|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 20.545 | 0 / 0 |
| 2 | 1 | 1 | 1 | 20.569 | 0 / 0 |
| 3 | 1 | 1 | 1 | 20.599 | 0 / 0 |
| 4 | 0 | 1 | 1 | 20.589 | 0 / 0 |
| 5 | 1 | 1 | 1 | 20.575 | 0 / 0 |
| 6 | 0 | 1 | 1 | 20.607 | 0 / 0 |
| 7 | 1 | 1 | 1 | 20.560 | 0 / 0 |
| 8 | 0 | 1 | 1 | 20.573 | 0 / 0 |
| 9 | 0 | 1 | 1 | 20.547 | 0 / 0 |
| 10 | 1 | 1 | 1 | 20.560 | 0 / 0 |

Target audit: 10 transitions; all recorded targets match: True. Terminal case: not_observed; time-truncation case: not_observed. A status of not_observed is not a pass. Separate software checks exercise both target branches and target-network freezing.

## MsPacman: training behavior




**Maximum Q.** Each point is the maximum over all visited states and all action outputs in that training episode. The online network changes during the episode. Q values are predictions, not measured game scores; a larger Q alone does not prove learning.

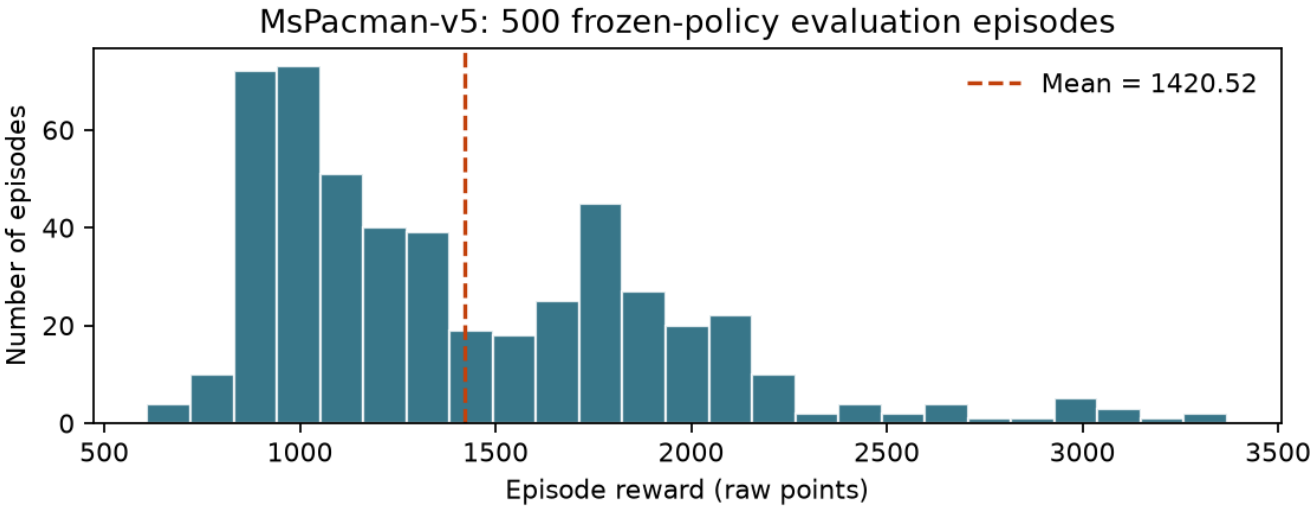



The first and final 100 training episodes average **257.00** and **1377.70** raw reward, respectively (difference +1120.70). The 100-episode moving average begins only when 100 complete episodes exist. Exploration changes over training, so the fixed-policy held-out comparisons on the next page provide a separate check of actual learned behavior.

Frozen evaluation uses the development-selected checkpoint from episode **1500**. The training curve includes exploratory actions and subsequent parameter updates; its final moving average therefore need not match the selected checkpoint's greedy evaluation score.

The last 5 development means range from 955.0 to 1583.0 points, ending at 1267.0. Late development scores rise and fall; the run does not demonstrate stable or monotonic convergence. Development scores select a checkpoint; the independent 500-episode result on the next page is the final performance estimate.

## MsPacman: frozen evaluation and learning validation




**Required 500-episode result:** mean 1420.52; sample standard deviation 511.69; range [610, 3370]. Evaluation uses epsilon=0 and seeds 100007 through 100506; no updates or replay sampling occur. Ending flags: 500 true terminations and 0 time truncations.

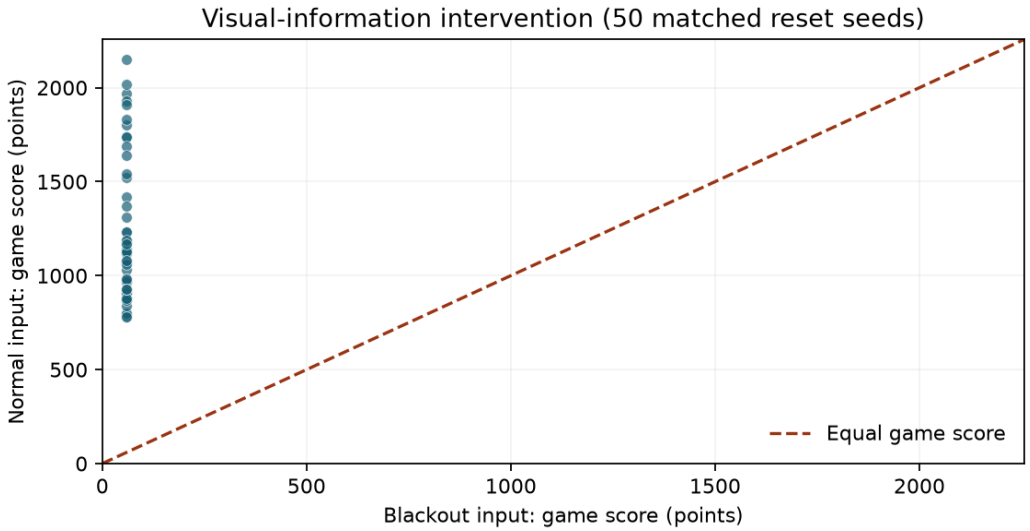
| Held-out policy | Episodes | Mean reward | Sample SD |
|---|---|---|---|
| Uniform random (epsilon=1) | 50 | 279.80 | 147.93 |
| Untrained network (epsilon=0) | 50 | 60.00 | 0.00 |
| Selected network (epsilon=0) | 50 | 1234.00 | 382.45 |

**Controlled learning check.** The three policies use the same 50 held-out reset seeds starting at 80000, separate from checkpoint selection and the required 500-episode evaluation. Initial and selected networks use identical greedy evaluation settings; the uniform-random control uses epsilon=1. Matched seeds do not imply identical trajectories once actions differ.

Selected minus random: mean paired difference 954.20 points; 95% paired-bootstrap interval [841.00, 1084.63] (n=50 pairs). The paired interval supports a positive improvement under this evaluation protocol.

Selected minus initial: mean paired difference 1174.00 points; 95% paired-bootstrap interval [1073.76, 1287.80] (n=50 pairs). The paired interval supports a positive improvement under this evaluation protocol.

# MsPacman: task-specific diagnostic and edge cases




**Nonstandard visual-information intervention.** Each point compares normal and all-zero network inputs under one matched reset seed. Above the diagonal, normal observations score higher. The environment, action set and raw score remain unchanged; the zero tensor removes visual information and also creates out-of-distribution inputs.

Normal minus blackout: mean paired difference 1174.00 points; 95% paired-bootstrap interval [1073.76, 1287.80] (n=50 pairs). The paired interval supports a positive improvement under this evaluation protocol.

**Auditable first ten steps.** The selected policy's validation rollout at seed 80000 records observations (or image summaries/hashes), every action value, selected action, raw/training rewards, target and ending flags in validation_first10_steps.json.

| Step | Action | Raw reward | Training reward | Target | Term / trunc |
|---|---|---|---|---|---|
| 1 | 8 | 0 | 0 | 2.855 | 0 / 0 |
| 2 | 8 | 0 | 0 | 2.890 | 0 / 0 |
| 3 | 5 | 0 | 0 | 2.895 | 0 / 0 |
| 4 | 8 | 0 | 0 | 2.910 | 0 / 0 |
| 5 | 6 | 0 | 0 | 2.915 | 0 / 0 |
| 6 | 1 | 0 | 0 | 2.913 | 0 / 0 |
| 7 | 1 | 0 | 0 | 2.917 | 0 / 0 |
| 8 | 5 | 0 | 0 | 2.933 | 0 / 0 |
| 9 | 5 | 0 | 0 | 2.936 | 0 / 0 |
| 10 | 8 | 0 | 0 | 2.883 | 0 / 0 |

Target audit: 10 transitions; all recorded targets match: True. Terminal case: not_observed; time-truncation case: not_observed. A status of not_observed is not a pass. Separate software checks exercise both target branches and target-network freezing.

# Reproducibility, validation and boundary handling

| Recorded compute | CartPole | MsPacman |
|---|---|---|
| hostname | mlds-deepdish3 | mlds-deepdish3 |
| gpu | NVIDIA GeForce RTX 2080 Ti | NVIDIA GeForce RTX 2080 Ti |
| device | cuda | cuda |
| cuda_visible_devices | 2 | 2 |
| python | 3.12.16 | 3.12.16 |
| torch | 2.14.1+cu126 | 2.14.1+cu126 |
| gymnasium | 1.4.0 | 1.4.0 |
| ale_py | 0.12.1 | 0.12.1 |

Run from the repository root with Python 3.12. The commands install the CUDA 12.6 PyTorch build used on deepdish before the remaining dependencies. On deepdish, set CUDA_VISIBLE_DEVICES to one available GPU only, then use --device cuda:0. The recorded visible-device setting above identifies the actual run. The saved config.json also includes source hashes, library versions and all hyperparameters.

```
python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install torch==2.14.1 --index-url https://download.pytorch.org/whl/cu126
python -m pip install -r requirements.txt
export CUDA_VISIBLE_DEVICES=2
python train.py --task cartpole --episodes 1500 --seed 7 --learning-rate 0.0005 \
  --eval-every 50 --eval-episodes 20 --epsilon-decay-steps 30000 \
  --device cuda:0 --output results/rerun/cartpole
python train.py --task mspacman --episodes 2000 --seed 7 --learning-rate 0.0001 \
  --eval-every 100 --eval-episodes 10 --epsilon-decay-steps 500000 \
  --device cuda:0 --output results/rerun/mspacman
python validate.py --task cartpole --run results/rerun/cartpole --episodes 50 \
  --grid-size 7 --grid-repeats 5 --device cuda:0
python validate.py --task mspacman --run results/rerun/mspacman --episodes 50 \
  --device cuda:0
python -m pytest --cov=dqn --cov=train --cov=validate --cov=build_report
python build_report.py --results results/rerun --output hw2_report_rerun.pdf \
  --repo-url https://github.com/ZikunZheng1129/mlds490-hw2
```

**What each validation establishes.** Random/initial/selected comparisons test learned control with fixed evaluation settings. The CartPole recovery map tests a broader initial-state distribution. MsPacman's input blackout tests whether visual observations affect behavior. The ten-step audit exposes the reward/target/action interaction. Paired bootstrap intervals use 2,000 resamples of matched episode seeds. These experiments supplement, rather than replace, unit/integration/end-to-end software tests.

**Boundary handling.** Learning waits for a full minibatch and warm-up; replay capacity is bounded and old records are replaced. Stored observations are copied, so later environment updates cannot corrupt replay. Invalid inputs and non-finite losses fail visibly. True terminals have reward-only targets; time limits retain bootstrap. MsPacman preserves full-game endings across life losses, nine actions and unsigned image storage. Short reward histories have no 100-episode average until the complete window exists. Checkpoints are saved atomically; evaluation freezes weights. Missing result files or fewer than 500 final episodes prevent report generation.

## Given an additional three months

**Month 1 - Repeatability and statistical precision.** Repeat each training setup with at least 10 independent training seeds and an identical interaction budget. Keep development and test seeds separate. Report per-run learning curves, score distributions, CartPole 500-step success rates and confidence intervals that include between-training-run variation. This directly addresses the largest limitation of the present single-run study.

**Month 2 - Controlled algorithm checks.** Change one factor at a time: remove replay, remove or vary target-network lag, change replay capacity, compare Huber to squared loss, and vary exploration schedules. Match environment decisions and evaluation rules across conditions. Measure final score, learning speed, TD-error distribution and instability. For MsPacman, compare one frame versus four and clipped versus raw learning rewards, while always evaluating raw full-game scores.

**Month 3 - Task-specific robustness and failure analysis.** Expand CartPole's starting-state grid and test labeled changes in pole length, cart mass and small observation noise. For MsPacman, compare controlled image occlusion, delayed observations and several sticky-action probabilities, preserving the standard v5 setup as the primary benchmark. Review failure trajectories and stratify results by state difficulty and game progress. Repeat apparent improvements on untouched test seeds before drawing conclusions.

**Interpretive limits.** A high predicted Q is not a direct correctness certificate. A positive paired interval is evidence for the tested frozen policies, not a guarantee about all retraining runs. The blackout intervention combines information loss with distribution shift, and the CartPole stress grid deliberately changes the reset distribution. Neither diagnostic replaces the required standard-environment results.

## References and accompanying evidence

[1] MLDS 490, Assignment 2 - Deep Q-Network; Week 6 DQN lecture; Lab Week 6. Course-provided requirements and algorithm reference.

[2] PyTorch, Reinforcement Learning (DQN) Tutorial. Neural action values, replay, Huber loss and detached targets.

[3] Farama Foundation, Arcade Learning Environment documentation. Atari environment versions, frame skipping and sticky actions.

[4] Farama Foundation, Handling Time Limits. Termination versus truncation for bootstrapping.

The repository includes dqn.py, train.py, validate.py, build_report.py, tests, requirements and README instructions. Each results directory preserves configuration, training/development records, the 500 evaluation scores, validation tables, first-ten-step trace and checkpoint files. Figures are regenerated from those records; building this PDF does not retrain models.